

An Assignment Manual for

Probability and Statistics

(BE03000251)

**B.E. Semester 3 (BM, Civil, CE, EC,
Environment Engineering, IT, Plastic
Technology)**



Government Engineering College, Rajkot.

**Directorate of Technical Education,
Gandhinagar, Gujarat**

Probability and Statistics

(BE03000251)

An Assignment Manual is prepared by

**Dr. Sunil G Sonchhatra
(Sciences and Hum. Dept.
& Assistant Professor in Mathematics)
GEC Rajkot**

&

**Dr. Nikhil D Abhangi
(Sciences and Hum. Dept.
& Assistant Professor in Mathematics)
GEC Rajkot**

Certificate

This is to certify that Mr./Ms. _____
_____ Enrollment No. _____ of B.E. Semester _____ Branch
_____ Engineering of this Institute (GTU Code: _____) has
satisfactorily completed the Assignment for the subject **Probability &
Statistics** for the academic year 2025-26.

Place: _____

Date: _____

Name and Sign of Faculty member

Preface

Main objectives of assignment work of any subject is for pleasing to the eye on required skills as well as creating ability amongst students to solve real time problem by developing relevant competencies in psychomotor domain. By keeping in view, GTU has designed competency focused outcome-based curriculum for engineering degree programs where sufficient time given to Assignment or Tutorial work. It shows importance of enhancement of skills amongst the students and it pays attention to utilize every second of time allotted for Assignment or Tutorial work amongst students, faculty members to achieve relevant outcomes by solving the Assignment or Tutorial rather than having simply study type classroom learning. It is must for effective implementation of competency focused outcome-based curriculum that every Assignment or Tutorial work is keenly designed to serve as a tool to develop and enhance relevant competency required by the various branch of engineering among every student. These psychomotor skills are very difficult to develop through traditional chalk and board content delivery method in the classroom. Accordingly, this lab manual is designed to focus on the programme and Course Outcome defined relevant programme, rather than old practice of conducting Assignment or Tutorial to prove concept and theory.

By using this Assignment or Tutorial manual students can go through the relevant theory and numerical in advance which creates an interest and students can have basic practice prior to exam. This in turn enhances pre-determined outcomes amongst students. Each Assignment or Tutorial in this manual begins with understanding, utility, modeling, analytic and creativity other relevant skills, course outcomes as well as practical outcomes (objectives).

This Assignment or Tutorial also provides guidelines to faculty members to facilitate student centric Assignment or Tutorial activities through each Assignment or Tutorial by arranging and managing necessary resources in order that the students follow the procedures with required skill to achieve the outcomes. It also gives an idea that how students will be assessed by process of continuous evaluation system.

Probability and Statistics is a basic science course, which focus on topics like, Experiment, definition of probability, conditional probability, independent events, Bayes' rule, Bernoulli trials, Random variables, discrete random variable, probability mass function, continuous random variable, probability density function, cumulative distribution function, properties of cumulative distribution function, Binomial distribution, Poisson distribution, Poisson approximation to the binomial distribution, Normal, Exponential and Gamma densities, Evaluation of statistical parameters for these distributions, Measure of central tendency: Moments, Expectation, dispersion, skewness, kurtosis, Linear Correlation, correlation coefficient, rank correlation coefficient, Regression, Formation of Hypothesis, Test of significance: Large sample test for single proportion, Difference of proportions, Single mean, Difference of means, and Difference of standard deviations, t- Test for single mean, difference of means, t-test for correlation coefficients, F- test for ratio of variances, Chi-square test for goodness of fit, Curve fitting by the numerical method: Curve fitting by of method of least squares, fitting of straight lines, second degree parabola and more general curves.

Probability & Statistics (BE03000251)													
CO-PO Matrix													
Table													
Sr. No.	Cos	Statement											
1	1	Define the basic probability terminology and also can describe random variables and their probability functions.											20%
2	2	Observe and analyze the behaviour of various discrete and continuous probability distribution functions											25%
3	3	Compute various statistical measures and further, calculate correlation and regression coefficient											20%
4	4	Apply the statistical methods for testing the significance of the large and small sample data by using large sample test, t- test, F- test and Chi-square test.											25%
5	5	Apply the least square method for curve fitting.											10%
Table CO – PO Matrix													
Sr. No.		PO1 Engineering knowledge	PO2 Problem analysis	PO3 Design/ development of solutions	PO4 Conduct investigations of complex problems	PO5 Modern tool usage	PO6 The engineer and society	PO7 Environment and sustainability	PO8 Ethics	PO9 Individual and team work	PO10 Communication	PO11 Project management and finance	PO12 Life-long learning
1	CO1	3	3										
2	CO2	3	3										
3	CO3	3	3										
4	CO4	3	3										
5	CO5	3	3										

Guidelines for Faculty members

1. Teacher should provide the guideline with demonstration some problems to the students.
2. Teacher shall explain in shortly, basic concepts/theory related to the Assignment or Tutorial to the students before starting of each Assignment or Tutorial
3. Involve all the students in Assignment or Tutorial.
4. Teacher is expected to share the skills and competencies to be developed in the students and ensure that the respective skills and competencies are developed in the students after the completion of the Assignment or Tutorial.
5. Teachers should give opportunity to students for hands-on experience after the demonstration.
6. Teacher may provide additional knowledge and skills to the students even though not covered in the manual but are expected from the students.
7. Give assignment and assess the performance of students based on task assigned to check whether it is as per the instructions or not.
8. Teacher is expected to refer complete curriculum of the course and follow the guidelines for implementation.

Instructions for Students

1. Students are expected to carefully listen to all the theory classes delivered by the faculty members and understand the Cos, content of the course, teaching and examination scheme, skill set to be developed etc.
2. Students shall organize the work in the group and make record of all work.
3. Students shall develop maintenance skill as expected by course.

4. Student shall attempt to develop related hand-on skills and build confidence.
5. Student shall develop the habits of evolving more ideas, innovations, skills etc. apart from those included in scope of manual.
6. Student shall refer book and resources.
7. Student should develop a habit of submitting the Assignment or Tutorial work as per the schedule and he/she should be well prepared for the same.

Index (Progressive Assessment Sheet)

Sr. No.	Objective(s) of Assignment	Page No.	Date of starting	Date of submission	Marks	Sign	Remarks
1	Calculation of probabilities using Classical Probability Theory,						
2	Apply Bayes' rule to calculate probability and also calculate probability distribution function by using probability mass function and probability density function						
3	Computation of Binomial and Poisson probabilities						
4	Computation of probabilities for Normal Distribution, Gamma distribution and exponential distribution						
5	Computation of measures of central tendency and moment, Expectation, dispersion, skewness for given data						
6	Compute various statistical measures and further, calculate correlation and regression coefficient						
7	Formation of hypothesis, test of significance: Large sample test for single proportion, difference of proportions, single mean, difference of means, and difference of standard deviations						
8	Test of significance for small samples: t- test for single mean, t-test for difference of means, t-test for correlation coefficients.						
9	Using Chi-square test and F- test to check significance of given data						
10	Apply the least square method for curve fitting like as straight line, second degree parabola and general curves.						
Total							

Date: _____	
Assignment/Tutorial No: 01	
Topic:	Basic Probability:
Sub Topics:	definition of probability, conditional probability, independent events,
Relevant CO:	Define the basic probability terminology and also can describe random variables and their probability functions.
Objectives:	1. Understand concept of Experiment, 2. Definition of probability, 3. Evaluate conditional probability 4. Define independent event. 5. Using probability theorems to find probability of particular problems.

Sr. No.	Question	CO	PI*	B.T. level
1	Define Mutually exclusive events and independent events. If A and B are independent events, where $P(A)=1/4$, $P(B)=2/3$, Find $P(A \cup B)$	Co1	1.2.2	R, U
2	A survey determines that in a locality, 33% go to work by Bike, 42% go by Car, and 12% use both. Find the probability that a random person selected uses neither of them.	Co1	1.2.2 2.8.1	U, A
3	In a high school graduating class of 100 students, 54 studied mathematics, 69 studied history and 35 studied both mathematics and history. If one of these students is selected at random, find the probability that (a) The students took mathematics or history. (b) The students did not take either of these subjects. (c) The students took history but not mathematics.	Co1	1.2.2 2.8.1	U, A
4	A problem is given to 5 students P, Q, R, S, T. If the probability of solving the problem individually is $1/2$, $1/3$, $2/3$, $1/5$, $1/6$ respectively, then find the probability that the at least one can solve this problem.	Co1	1.2.2 2.8.1	U, A
5	A and B take turns in throwing two dice, the first to throw 9 being awarded the first prize. Show that if A has the first throw, their chance of winning are in the ratio 9:8.	Co1	1.2.2 2.8.1	U, A
6	A speaks truth in 75% cases and B in 80% cases. Find the probability that they are likely to contradict each other in stating the same fact.	Co1	1.2.2 2.8.1	U, A
7	There are two boxes A and B containing 4 white, 3 red and 3 white, 7 red balls respectively. A box is chosen at random and a ball is drawn from it. If the ball is white, find the probability that it is from box A.	CO1	1.2.2 2.8.1	U, A
8	If 3 balls are “randomly drawn” from a bowl containing 6 white and 5 black balls, what is the probability that one of the balls is white and the other two black?	CO1	1.2.2 2.8.1	U, A
9	Suppose box A contains 4 red and 5 blue coins and box B contains 6 red and 3 blue coins. A coin is chosen at random from the box A and placed in box B. Finally, a coin is chosen at random from among that now in box B. What is the probability a blue coin was transferred from box A to box B given that the coin chosen from box B is red?	CO1	1.2.2 2.8.1	U, A
10	If 3 balls are “randomly drawn” from a bowl containing 6 white and 5 black balls, find the probability that one of the balls is white and the other two are black.	CO1	1.2.2 2.8.1	U, A

B.T. Level: R: Remembering, U: Understanding, A: Applying, N: Analyzing, E: Evaluating, C: Creating

PI* <https://www.aicte-india.org/sites/default/files/ExaminationReforms.pdf>

Date:	
Assignment No: 02	
Topic:	Random Variables:
Sub Topics:	Bayes' rule, Bernoulli trials, discrete random variable, probability mass function, continuous random variable, probability density function, cumulative distribution function, properties of cumulative distribution function.
Relevant CO:	Define the basic probability terminology and also can describe random variables and their probability functions
Objectives:	1. Define Random variables 2. Define discrete random variable and Continuous random variable. 3. Use probability mass function and probability density function to calculate probability of various problems. 4. Define Cumulative distribution function and also properties of cumulative distribution function

Sr. No.	Question	CO	PI	B.T. level
1	State Bayes' theorem. A company selected engineers through campus interview from a university. Out of the total selection made 60%, 30% and 10% are from Electronics, Computer and Mechanical respectively. If 9%, 20% and 60% of the selected students do not join the company, what is the probability that they are from computer stream?	CO1	1.2.2 2.8.1	R, U , A
2	Two boxes containing candies are placed on a table. The boxes are labeled B1 and B2. Box B1 contains 7 cinnamon candies and 4 ginger candies. Box B2 contains 3 cinnamon candies and 8 pepper candies. The boxes are arranged so that the probability of selecting box B1 is $\frac{1}{3}$ and the probability of selecting box B2 is $\frac{2}{3}$. Suresh is blindfolded and asked to select a candy. He will win a colour TV if he selects a cinnamon candy. What is the probability that Suresh will win the TV (that is, he will select a cinnamon candy)?	CO1	1.2.2 2.8.1	U, A
3	For a certain binary Communication channel, the probability that a transmitted 'O' is received as 'O' is 0.95 and the probability that a transmitted '1' is received as '1' is 0.90. If the probability that a 'O' is transmitted is 0.4, find the probability that (a) a '1' is received . (b) a '1' is transmitted given that a '1' is received .	CO1	1.2.2 2.8.1	U, A
4	At a certain university, 4% of men are over 6 feet tall and 1% of women are over 6 feet tall. The total student population is divided in the ratio 3:2 in favor of women. If a student is selected at random from among all those over six feet tall, what is the probability that the student is a woman?	CO1	1.2.2 2.8.1	U, A
5	Three companies A, B and C supply 25%, 35% and 40% of the notebooks to a school. Past experience shows that 5%, 4% and 2% of the notebooks produced by these companies are defective. If a notebook was found to be defective, what is the probability that the notebook was supplied by A?	CO1	1.2.2 2.8.1	U, A
6	An insurance company insured 2000 bike drivers, 4000 car drivers and 6000 truck drivers. The probability of an accident involving a bike driver, a car driver and a truck driver is 0.10, 0.03 and 0.15 respectively. One of the insured persons meets with an accident. What is the probability that he is a bike driver?	CO1	1.2.2 2.8.1	R, U , A

7	Define a term random variable and explain different types of random variable.	CO1	1.2.2	R																				
8	If X is random variable denoting the sum obtained in rolling a pair of fair dice, determine its probability mass function.	CO1	1.2.2 2.8.1	U , A																				
9	Find the distribution function for the random variable X where probability density is given by $f(x)=\begin{cases} \frac{1}{2}(x+1) ; \text{ for } -1 \leq x \leq 1 \\ 0, \text{ otherwise} \end{cases}$	CO1	1.2.2 2.8.1	U , A																				
10	A random variable X has the following probability distribution: (a) <table border="1"><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>p(x)</td><td>0</td><td>k</td><td>2k</td><td>2k</td><td>3k</td><td>k²</td><td>2k²</td><td>7k²+k</td></tr></table> n Find k and hence evaluate P(0 < X < 5) (b) Find $P\left(\frac{1.5 < X < 4.5}{X > 2}\right)$ (c) Find E(X).	x	0	1	2	3	4	5	6	7	p(x)	0	k	2k	2k	3k	k ²	2k ²	7k ² +k	CO1	1.2.2 2.8.1	U , A		
x	0	1	2	3	4	5	6	7																
p(x)	0	k	2k	2k	3k	k ²	2k ²	7k ² +k																
11	If X is a continuous random variable with the following probability density function $f(x)=\begin{cases} a(2x-x^2), 0 < x \leq 2 \\ 0, x > 2 \end{cases}$ (i) Find the value of "a". (ii) Find P[X >1].	CO1	1.2.2 2.8.1	U , A																				
12	A random variable X has the following probability distribution <table border="1"><tr><td>Values of X</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr><tr><td>p(x)</td><td>a</td><td>3a</td><td>5a</td><td>7a</td><td>9a</td><td>11a</td><td>13a</td><td>15a</td><td>17a</td></tr></table> (i) Determine the value of a. (ii) Find P(x<3); p(X ≥ 3), p (0<X<5). (iii) What is the smallest value of x for which P(X < x) > 0.5. (iv) Find E(X).	Values of X	0	1	2	3	4	5	6	7	8	p(x)	a	3a	5a	7a	9a	11a	13a	15a	17a	CO1	1.2.2 2.8.1	U , A
Values of X	0	1	2	3	4	5	6	7	8															
p(x)	a	3a	5a	7a	9a	11a	13a	15a	17a															

B.T. Level: R: Remembering, U: Understanding, A: Applying, N: Analyzing, E: Evaluating, C: Creating

Date:	
Assignment No: 03	
Topic:	Some special Probability Distributions:
Sub Topics:	Binomial distribution, Poisson distribution, Poisson approximation to the binomial distribution
Relevant CO:	Observe and analyze the behaviour of various discrete and continuous probability distributions
Objectives:	1. Understand concept of discrete probability distribution 2. Definition of Binomial distribution 3. Evaluation of statistical parameters for binomial distribution 4. Definition of Poisson distribution 5. To prove Poisson approximation to the binomial distribution 6. Evaluation of statistical parameters for Poisson distribution

Sr. No.	Question	CO	PI	B.T. level																								
1	If 10% of the pens manufactured by the company are defective, find the probability that a box of 12 pens contain (i) Exactly two defective pens (ii) At least two defective pens (iii) No defective pens.	CO2	1.2.2 2.8.1	U , A																								
2	If During a war, one out of 9 ships could not arrive safely. Find the probability that exactly 3 out of 6 would arrive safely.	CO2	1.2.2 2.8.1	U , A																								
3	A noisy transmission channel has per digit error probability $p = 0.01$, Find probability of more than one error in 10 received digit using (i) Binomial distribution (ii) Poisson distribution	CO2	1.2.2 2.8.1	U , A																								
4	Letters are received in an office on each one of 100 days. Assuming the following data to form a random sample from a Poisson distribution, find the expected frequencies correct to nearest units. (Given $e^{-4} = 0.0183$) <table border="1"><tr><td>No. of letters (X)</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr><tr><td>Frequency(f)</td><td>1</td><td>4</td><td>15</td><td>22</td><td>21</td><td>20</td><td>8</td><td>6</td><td>2</td><td>0</td><td>1</td></tr></table>	No. of letters (X)	0	1	2	3	4	5	6	7	8	9	10	Frequency(f)	1	4	15	22	21	20	8	6	2	0	1	CO2	1.2.2 2.8.1	U , A
No. of letters (X)	0	1	2	3	4	5	6	7	8	9	10																	
Frequency(f)	1	4	15	22	21	20	8	6	2	0	1																	
5	Out of 800 families with 4 children each, how many families would be expected to have (i) no girl (ii) at least one boy (iii) 2 boys and 2 girls (iv) at most 2 girls Assume equal probabilities for boys and girls .	CO2	1.2.2 2.8.1	U , A																								
6	Razor blades are supplied by a manufacturing company in packets of 10. There is a probability of 1 in 100 blades to be defective. Use Poisson distribution to calculate the number of packets containing one defective blade, no defective blade and all defective in a consignment of 10,000 packets.	CO2	1.2.2 2.8.1	U , A																								
7	Each sample of a Chemical used in a textile dying process has a 10% chance of containing a pollutant. Find out the probability that in the next 20 samples, contain the pollutant. Assuming that the samples are independent with regard to the presence of the pollutant. To determine the probability that at least four sample contain the pollutant.	CO2	1.2.2 2.8.1	U , A																								
8	The probability that a bomb dropped from a plane will strike the target is 1/5. If six such bombs are dropped, find the probability that at least two will strike the target.	CO2	1.2.2 2.8.1	U , A																								
9	Find the probability that at most 5 defective fuses will be found	CO2	1.2.2	U , A																								

	in a box of 200 fuses if experience shows that 2 percent of such fuses are defective.		2.8.1	
10	With usual notations, find the value of p for a binomial random variable X when n=6 and $9P(X=4)=P(X=2)$.	CO2	1.2.2 2.8.1	U , A
11	Prove that Poisson distribution is a limiting case of Binomial distribution.	CO2	1.2.2 2.8.1	U , A

B.T. Level: R: Remembering, U: Understanding, A: Applying, N: Analyzing, E: Evaluating, C: Creating

Date:	
Assignment No: 04	
Topic:	Some special Probability Distributions:
Sub Topics:	Normal distribution, Exponential and Gamma densities, Evaluation of statistical parameters for these distributions.
Relevant CO:	Observe and analyze the behavior of various discrete and continuous probability distributions
Objectives:	1. Definition Normal distribution 2. Evaluation of statistical parameters for normal distribution 3. Definition of Exponential distribution 4. Evaluation of statistical parameters for Exponential distribution 5. Definition of Gamma distribution 6. Evaluation of statistical parameters for Gamma distribution

Sr. No.	Question	CO	PI	B.T. level
1	State properties of the normal distribution. Suppose the marks of 800 students are normally distributed with mean 66 and standard deviation 5, find the number of students getting marks (i) between 65 and 70 (ii) greater than or equal to 72.	CO2	1.2.2 2.8.1	R,U, A
2	Find the area under the normal curve for (i) $z < -1.8$ (ii) $z > 0.5$ (iii) $z < 2.3$	CO2	1.2.2 2.8.1	U , A
3	If Skulls are classified as A, B, C according as the length and breadth index as under 75, between 75 and 80, or over 80, find approximately the mean and the standard deviation of the classes in which A are 58% , B are 38% and C are 4% given that $f(t) = \frac{1}{\sqrt{2\pi}} \int_0^t e^{-t^2/2} dt \quad f(0.20) = 0.08 \text{ and } f(1.75) = 0.46 .$	CO2	1.2.2 2.8.1	U , A
4	A manufacturer knows from experience that the resistance of resistors he produces is normal with a mean $\mu = 100$ ohms and standard deviation $\sigma = 2$ ohms. What percentage of resistors will have resistance between 98 ohms and 102 ohms?	CO2	1.2.2 2.8.1	U , A
5	In a normal distribution, 31% of the items are under 45 and 8% are over 64. Find the parameters of the distribution.	CO2	1.2.2 2.8.1	U , A
6	Assume that the length of a phone call in minutes is an exponential with parameter $\lambda = \frac{1}{10}$. If someone arrives at a phone booth just before you arrive, Find the probability that you will have to wait (i) Less than 5 minutes (ii) between 5 and 10 minutes.	CO2	1.2.2 2.8.1	U , A

7	The time required to repair a machine is exponentially distributed with parameter $\lambda = \frac{1}{2}$. (i) What is the probability that the repair time exceed 2 hrs? (ii) What is the conditional probability that a repair takes at least 10 hrs? Given that its duration exceeds 9 hrs.	CO2	1.2.2 2.8.1	U , A
8	How exponential distribution is useful in real applications? Find the mean and variance of the exponential distribution $f(x) = \begin{cases} \frac{1}{\beta} e^{-x/\beta}, & x > 0, \quad \beta > 0. \\ 0, & \text{otherwise} \end{cases}$	CO2	1.2.2 2.8.1	U , A
9	Define Gamma distribution. Given a Gamma random variable X with $r = 3$ and $\lambda = 2$. Compute $E(X)$, $\text{Var}(X)$ and $P(X \leq 1.5)$.	CO2	1.2.2 2.8.1	U , A
10	If a random variable X is Gamma distribution with parameter $\lambda = 3$, compute the value of (i) $P(X \leq 1)$ and (ii) $P(1 \leq X \leq 2)$.	CO2	1.2.2 2.8.1	U , A
11	In a certain city, the daily consumption of an electric power (in millions of kilowatt hours) can be treated as a random variable having a gamma distribution with $\alpha = 2$ and $\beta = 3$. If the power plant of this city has a daily capacity of 12 million kilowatt-hours, what is the probability that this power supply will be inadequate on any given day? Also, find the mean of this probability density.	CO2	1.2.2 2.8.1	U , A

B.T. Level: R: Remembering, U: Understanding, A: Applying, N: Analyzing, E: Evaluating, C: Creating

Date:	
Assignment No: 05	
Topic:	Basic Statistics:
Sub Topics:	Measure of central tendency, mean , mode, median, mean deviation, standard deviation and quartile deviation, moment, Expectation, dispersion, skewness
Relevant CO:	Compute various statistical measures and further, calculate correlation and regression coefficient
Objectives:	1. Measure of central tendency 2. Find Mean, Mode and Median for given data. 3. Calculate mean deviation, standard deviation and quartile deviation. 4. Find Expectation for the given data. 5. Find dispersion and skewness for given data.

Sr. No.	Question	CO	PI	B.T. level																						
1	Find the mean, median and mode for the following frequency distribution <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr><tr><td>f</td><td>4</td><td>7</td><td>8</td><td>10</td><td>6</td><td>6</td><td>4</td><td>2</td><td>2</td><td>1</td></tr></table>	x	1	2	3	4	5	6	7	8	9	10	f	4	7	8	10	6	6	4	2	2	1	CO3	1.2.2 2.8.1	U, A
x	1	2	3	4	5	6	7	8	9	10																
f	4	7	8	10	6	6	4	2	2	1																
2	An insurance company obtained the following data for accident claims (in thousand rupees) from a particular region. Find its mean, median and mode. <table><tr><td>Amount</td><td>1-3</td><td>3-5</td><td>5-7</td><td>7-9</td><td>9-11</td><td>11-13</td></tr><tr><td>Frequency</td><td>6</td><td>47</td><td>75</td><td>46</td><td>18</td><td>8</td></tr></table>	Amount	1-3	3-5	5-7	7-9	9-11	11-13	Frequency	6	47	75	46	18	8	CO3	1.2.2 2.8.1	U , A								
Amount	1-3	3-5	5-7	7-9	9-11	11-13																				
Frequency	6	47	75	46	18	8																				
3	Compute mean deviation and standard deviation for the data	CO3	1.2.2	U ,																						

		Size of the item	6	7	8	9	10	11	12			2.8.1	A
		frequency	3	6	9	13	8	5	4				
4	Find the quartile deviation and mean deviation for the following distribution.										CO3	1.2.2 2.8.1	U , A
	Class	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80				
	frequency	2	5	7	13	21	16	8	3				
5	The first four moments of a distribution about the value 4 of the variable are -1.5, 17, -30 and 108. Find the moments about the mean β_1, β_2 and comment on their values. Also find the moment about the point $x = 2$.										CO3		
6	Find the first four moments (a) about the origin (b) about the mean for a random variable X having density function $f(x) = \begin{cases} \frac{4x(9-x^2)}{81}; & 0 \leq x \leq 3 \\ 0 & ; \text{otherwise} \end{cases}$										CO3	1.2.2 2.8.1	U , A
7	Calculate the first four moments of the following distribution about $a = 40.5$ and hence find the moments about the mean of distribution:										CO3	1.2.2 2.8.1	U , A
	Hours worked	30-32.9	33-35.9	36-38.9	39-41.9	42-44.9	45-47.9						
	No. of Industries	2	4	26	47	15	6						
8	A man draws 2 balls from a bag containing 3 white and 5 black balls. If he is to receive Rs. 50 for every white ball he draws and Rs. 20 for every black ball, find his expectation.										CO3	1.2.2 2.8.1	U , A
9	A random variable X has a following p. d. f. $\begin{cases} \frac{1}{2}x; & 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$ Find (i) $E(X)$ (ii) $V(x)$ (iii) $E(3X^2 - 2X)$										CO3	1.2.2 2.8.1	U , A
10	For a group of 10 items, $\sum x = 452$, $\sum x^2 = 24270$ and $\text{mode} = 43.7$. Find Karl Pearson's coefficient of skewness.										CO3	1.2.2 2.8.1	U , A

B.T. Level: R: Remembering, U: Understanding, A: Applying, N: Analyzing, E: Evaluating, C: Creating

Date:	
Assignment No: 06	
Topic:	Basic Statistics:
Sub Topics:	kurtosis, Linear Correlation, correlation coefficient, rank correlation coefficient, Regression
Relevant CO:	Compute various statistical measures and further, calculate correlation and regression coefficient
Objectives:	- Using moment to calculate skewness and kurtosis. - Calculate Correlation and Rank Correlation for given data. - Find regression line for given data.

Sr. No.	Question	CO	PI	B.T. level
1	For a distribution of 250 heights, calculations showed that the mean, standard deviation, β_1 and β_2 were 54 inches, 3 inches, 0 inches and 3 inches. It was however, discovered on checking that two items 64 and 50 in the original data were wrongly written in place of the correct	Co3	1.2.2 2.8.1	U , A

	values 62 and 52 inches respectively. Calculate the correct frequency constant.																													
2	Calculate the Karl Pearson's coefficient of correlation for the following data: <table><tr><td>X</td><td>25</td><td>27</td><td>30</td><td>35</td><td>33</td><td>28</td><td>36</td></tr><tr><td>Y</td><td>19</td><td>22</td><td>27</td><td>28</td><td>30</td><td>23</td><td>28</td></tr></table>	X	25	27	30	35	33	28	36	Y	19	22	27	28	30	23	28	Co3	1.2.2 2.8.1	U, A										
X	25	27	30	35	33	28	36																							
Y	19	22	27	28	30	23	28																							
3	Calculate the correlation coefficient for the following heights (in inches) of fathers (X) and their son(Y): <table><tr><td>X</td><td>65</td><td>66</td><td>67</td><td>67</td><td>68</td><td>69</td><td>70</td><td>72</td></tr><tr><td>Y</td><td>67</td><td>68</td><td>65</td><td>68</td><td>72</td><td>72</td><td>69</td><td>71</td></tr></table>	X	65	66	67	67	68	69	70	72	Y	67	68	65	68	72	72	69	71	Co3	1.2.2 2.8.1	U, A								
X	65	66	67	67	68	69	70	72																						
Y	67	68	65	68	72	72	69	71																						
4	Prove that the correlation coefficient lies between – 1 and + 1.	Co3	1.2.2 2.8.1	U, A																										
5	Calculate the rank correlation coefficient for given data <table><tr><td>X</td><td>12</td><td>15</td><td>18</td><td>20</td><td>16</td><td>15</td><td>18</td><td>22</td><td>15</td><td>21</td><td>18</td><td>15</td></tr><tr><td>Y</td><td>10</td><td>18</td><td>19</td><td>12</td><td>15</td><td>19</td><td>17</td><td>19</td><td>16</td><td>14</td><td>13</td><td>17</td></tr></table>	X	12	15	18	20	16	15	18	22	15	21	18	15	Y	10	18	19	12	15	19	17	19	16	14	13	17	Co3	1.2.2 2.8.1	U, A
X	12	15	18	20	16	15	18	22	15	21	18	15																		
Y	10	18	19	12	15	19	17	19	16	14	13	17																		
6	The ranks of same 16 students in Mathematics and Physics are as follows. Two numbers within brackets denote the ranks of the students in Mathematics and Physics. (1,1), (2,10), (3,3), (4,4), (5,5), (6,7), (7,2), (8,6), (9,8), (10,11), (11,15), (12,9), (13,14), (14,12), (15,16), (16,13). Calculate the rank correlation coefficient for proficiencies of this group in Mathematics and Physics	Co3	1.2.2 2.8.1	U, A																										
7	Calculate the coefficient of correlation and obtain the line of regression for the following data. <table><tr><td>X</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td></tr><tr><td>y</td><td>9</td><td>8</td><td>10</td><td>12</td><td>11</td><td>13</td><td>14</td><td>16</td><td>15</td></tr></table> Obtain also an estimate for y which should correspond on an average to x = 6.2.	X	1	2	3	4	5	6	7	8	9	y	9	8	10	12	11	13	14	16	15	Co3	1.2.2 2.8.1	U, A						
X	1	2	3	4	5	6	7	8	9																					
y	9	8	10	12	11	13	14	16	15																					
8	In a partially destroyed laboratory on record of an analysis of correlation data, the following results only are legible, Var x = 9, Regression equations : 8x – 10y + 66 = 0, 40x – 18 y =214 Find : (i) The mean of x and y (ii) The standard deviation of y (iii)The coefficient of correlation between x and y .	Co3	1.2.2 2.8.1	U, A																										
9	Show that θ , the acute angle between the two lines of regression is given by $\tan \theta = \left(\frac{1-r^2}{r} \right) \left(\frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} \right)$ Interpret the case when r = 0, ± 1	Co3	1.2.2 2.8.1	U, A																										
10	Find the most likely price in Bombay corresponding to the price of Rs.70 at Calcutta from the following: <table><tr><td></td><td>Calcutta</td><td>Bombay</td></tr><tr><td>Average price</td><td>65</td><td>67</td></tr><tr><td>Standard deviation</td><td>2.5</td><td>3.5</td></tr></table> Correlation coefficient between the prices of commodities in the two cities is 0.8.		Calcutta	Bombay	Average price	65	67	Standard deviation	2.5	3.5	Co3	1.2.2 2.8.1	U, A																	
	Calcutta	Bombay																												
Average price	65	67																												
Standard deviation	2.5	3.5																												

B.T. Level: R: Remembering, U: Understanding, A: Applying, N: Analyzing, E: Evaluating, C: Creating

Date:	
Assignment No: 7	
Topic:	APPLIED STATISTICS
Sub Topics:	Formation of Hypothesis, Test of significance: Large sample Test for single proportion, Difference of proportions, Single mean, Difference of means, and Difference of standard deviations
Relevant CO:	Apply the statistical methods for testing the significance of the large and small sample data by using large sample test, t- test, F- test and Chi-square test.
Objectives:	1. Understand concept of Hypothesis 2. Test of significance for Large sample for single proportion & difference of proportions 3. Test of significance for Large sample for single mean & difference of means 4. Test of significance for Large sample for difference of standard deviations

Sr. No.	Question	CO	PI	B.T. level												
1	Explain the term related to testing of hypothesis (i) Null hypothesis (ii) Level of significance (iii) Two tailed test.	Co4	1.2.2	R												
2	In a random sample of 400 persons from a large population, 120 are females. Can it be said that males and females are in the ratio 5:3 in population? Use 1% level of significance.	Co4	1.2.2 2.8.4	U, A												
3	Twenty people were attacked by disease and only 18 survived. Will you reject the hypothesis that the survival rate, if attacked by this disease is 85% in favor of the hypothesis that it is more at 5% level. (use large sample Test)	Co4	1.2.2 2.8.4	U, A												
4	A cigarette manufacturing firm claims that its brand A of the cigarettes outsells its brand B by 8%. If it is found that 42 out of a sample of 200 smokers prefer brand A and 18 out of another random sample of 100 smokers prefer brand B, Test whether the 8% difference is a valid claim. (use 5% LOS)	Co4	1.2.2 2.8.4	U, A												
5	In a large population, there are 30% and 25% respectively of blue eyed people. Is this difference likely to be hidden in samples of 1200 and 900 respectively from the two populations.	Co4	1.2.2 2.8.4	U , A												
6	Before an increase in excise duty on tea, 800 persons out of a sample of 1000 persons were found to be tea drinkers. After an increase in duty, 800 people were tea drinkers in sample of 1200. Using standard error of proportion, state whether there is a significant decrease in the consumption of tea after the increase in excise duty.	Co4	1.2.2 2.8.4	U, A												
7	The mean breaking strength of the cables supplied by a manufacturer is 1800 with SD of 100. By a new technique in the manufacturing process, it is claimed that the breaking strength of the cable has increased. To Test this claim, a sample of 50 cables is Tested and it is found that the mean breaking strength is 1850. Can we support the claim at 1% LOS.	Co4	1.2.2 2.8.4	U, A												
8	A normal population has mean of 6.8 and standard deviation of 1.5. A sample of 400 members gave a mean of 6.75. Is the difference significant at LOS 5%. (large sample Test)	Co4	1.2.2 2.8.4	U, A												
9	Intelligence Tests were given to groups of boys and girls . <table><tr><td></td><td>Mean</td><td>Standard Deviation</td><td>Size</td></tr><tr><td>Girls</td><td>75</td><td>8</td><td>60</td></tr><tr><td>Boys</td><td>73</td><td>10</td><td>100</td></tr></table>		Mean	Standard Deviation	Size	Girls	75	8	60	Boys	73	10	100	Co4	1.2.2 2.8.4	U, A
	Mean	Standard Deviation	Size													
Girls	75	8	60													
Boys	73	10	100													

	Examine if the difference between mean scores is significant at 5% LOS.															
10	Random sample drawn from two countries gave the following data relating to the heights of adult males: <table border="1"><thead><tr><th></th><th>Mean</th><th>Standard Deviation</th><th>Size</th></tr></thead><tbody><tr><td>A</td><td>67.42</td><td>2.58</td><td>1000</td></tr><tr><td>B</td><td>67.25</td><td>2.50</td><td>1200</td></tr></tbody></table> (i) Is the difference between means significant? (ii) Is the difference between standard deviations significant? (At 5% LOS)		Mean	Standard Deviation	Size	A	67.42	2.58	1000	B	67.25	2.50	1200	Co4	1.2.2 2.8.4	U, A
	Mean	Standard Deviation	Size													
A	67.42	2.58	1000													
B	67.25	2.50	1200													
11	A tire company is suspicious to claim that the average lifetime of certain tires is at least 28000 km. To check the claim, the company takes the sample of 40 tires and gets a mean life time of 27463 km with standard deviation of 1348 km. Test the hypothesis at 1% level of significance.	Co4	1.2.2 2.8.4	U, A												
12	A stenographer claims that she can type at the rate of 120 words per minute. She demonstrated, on the basis of 100 trials, an average speed of 116 words with a standard deviation of 15 words. Does this enable us to reject the null hypothesis $\mu = 120$ against the alternative hypothesis $\mu < 120$ at the 0.05 level of significance?	Co4	1.2.2 2.8.4	U, A												

B.T. Level: R: Remembering, U: Understanding, A: Applying, N: Analyzing, E: Evaluating, C: Creating

Date:	
Assignment No: 8	
Topic:	APPLIED STATISTICS
Sub Topics:	Test of significance for small samples: t- test for single mean, t-test for difference of means, t-test for correlation coefficients.
Relevant CO:	Apply the statistical methods for testing the significance of the large and small sample data by using large sample test, t- test, F- test and Chi-square test.
Objectives:	1. Test of significance for small samples using t- test for single mean 2. Using t-test for difference of means, t-test for correlation coefficient

Sr. No.	Question	CO	PI	B.T. level																
1	A mechanist is making engine parts with axle diameters 0.700”. A random sample of 10 parts shows a mean diameter of 0.742” with a SD of 0.04”. Test whether work is meeting the specification.	Co4	1.2.2 2.8.4	U, A																
2	Two horses A and B were tested according to the time (in seconds) to run a particular track with the following results: <table border="1"><tr><td>Horse A :</td><td>28</td><td>30</td><td>32</td><td>33</td><td>33</td><td>29</td><td>34</td></tr><tr><td>Horse B :</td><td>29</td><td>30</td><td>30</td><td>24</td><td>27</td><td>29</td><td></td></tr></table> Test whether you can discriminate between the two horses. You can use the fact that 5% value of t for 11 d. f. is 2.20.	Horse A :	28	30	32	33	33	29	34	Horse B :	29	30	30	24	27	29		Co4	1.2.2 2.8.4	U, A
Horse A :	28	30	32	33	33	29	34													
Horse B :	29	30	30	24	27	29														
3	For a random sample of 10 pigs, fed the diet A, the increase in weight in a certain period were 10, 6, 16, 17, 13, 12, 8, 14, 15, 9 lbs. For another random sample of 12 pigs, fed on diet B, the increases in the same period were 7, 13, 22, 15, 12, 14, 18, 8, 21, 23, 10, 17 lbs.	Co4	1.2.2 2.8.4	U, A																

	These whether the diets A and B differ significantly as regards the effect on increase in weight (or Test whether the mean increases in the two samples as significantly different).																														
4	In a certain experiment to compare two types of pig-foods A and B, the following results of increase in weight were observed in pigs : <table border="1"><tr><td>Pig No.</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr><tr><td>Wt. in kg. Food A</td><td>49</td><td>53</td><td>51</td><td>52</td><td>47</td><td>50</td><td>52</td><td>53</td></tr><tr><td>Wt. in kg. Food B</td><td>52</td><td>55</td><td>52</td><td>53</td><td>50</td><td>54</td><td>54</td><td>53</td></tr></table> (a) Assuming that two samples of pigs are independent, can we conclude that food B is better than food A? (b) Examine the case when the same set of eight pigs were used in both the foods.	Pig No.	1	2	3	4	5	6	7	8	Wt. in kg. Food A	49	53	51	52	47	50	52	53	Wt. in kg. Food B	52	55	52	53	50	54	54	53	Co4	1.2.2 2.8.4	U, A
Pig No.	1	2	3	4	5	6	7	8																							
Wt. in kg. Food A	49	53	51	52	47	50	52	53																							
Wt. in kg. Food B	52	55	52	53	50	54	54	53																							
5	A random sample of size 15 from bivariate normal distribution gave a correlation coefficient $r=0.5$. Does this indicate the existence of correlation in the population?	Co4	1.2.2 2.8.4	U, A																											
6	A random sample from a company's very extensive files shows that orders for a certain piece of machinery were filled, respectively, in 10, 12, 19, 14, 15, 18, 11 and 13 days. Use the level of significance $\alpha = 0.01$, to test the claim that on average such orders are filled in 10.5 days. Choose the alternative hypothesis so that rejection of the null hypothesis $\mu = 10.5$ implies that it takes longer than indicated. Assume normality.	Co4	1.2.2 2.8.4	U, A																											
7	Manager of a diet plan advertise that the mean weight loss for people on their plan is at least 45 pounds in 6 months. A sample of 28 people on this plan loses an average of 35 pounds with standard deviation of 20 pounds. Test at 1% level of significance.	Co4	1.2.2 2.8.4	U, A																											
8	It is known that the mean diameter of rivets produced by two firms A and B were practically same but their SDs may differ. For 22 rivets manufactured by B, the SD is 3.8. Test whether the products of A have the same variability as those of B.	Co4	1.2.2 2.8.4	U, A																											
9	The nicotine contents in two random samples of tobacco are given below : <table border="1"><tr><td>Sample 1 :</td><td>21</td><td>24</td><td>25</td><td>26</td><td>27</td><td></td></tr><tr><td>Sample 2 :</td><td>22</td><td>27</td><td>28</td><td>30</td><td>31</td><td>36</td></tr></table> Can you say that the two samples came from the same population?	Sample 1 :	21	24	25	26	27		Sample 2 :	22	27	28	30	31	36	Co4	1.2.2 2.8.4	U , A													
Sample 1 :	21	24	25	26	27																										
Sample 2 :	22	27	28	30	31	36																									
10	A random sample of fifteen paired observations from a bivariate normal population gives a correlation coefficient of -0.5. Does this indicate existence of correlation in the sampled population? (Test at 5% level of significance)	Co4	1.2.2 2.8.4	U , A																											

B.T. Level: R: Remembering, U: Understanding, A: Applying, N: Analyzing, E: Evaluating, C: Creating

Date:	
Assignment No: 9	
Topic:	APPLIED STATISTICS
Sub Topics:	Chi-square Test and F- Test
Relevant CO:	Apply the statistical methods for testing the significance of the large and small sample data by using large sample test, t- test, F- test and Chi-square test.
Objectives:	- Using Chi-square test for goodness of fit. Testing the significance by using F-test.

Sr. No.	Question	CO	PI	B.T. level																				
1	The following table shows the number of plants having certain characters. <table><tr><td></td><td>Flat leaves</td><td>Curled leaves</td><td>Total</td></tr><tr><td>White flowers</td><td>267</td><td>27</td><td>294</td></tr><tr><td>Red flowers</td><td>757</td><td>155</td><td>912</td></tr><tr><td>Total</td><td>1024</td><td>182</td><td>1206</td></tr></table> Use the χ^2 test to access the correctness of the hypothesis that the flower colour is independent of flatness of leaf at 5% level of significance.		Flat leaves	Curled leaves	Total	White flowers	267	27	294	Red flowers	757	155	912	Total	1024	182	1206	Co4	1.2.2 2.8.4	U, A				
	Flat leaves	Curled leaves	Total																					
White flowers	267	27	294																					
Red flowers	757	155	912																					
Total	1024	182	1206																					
2	The mean life of a sample of 25 fluorescent light bulbs is found as 1550 hours with a standard deviation of 120 hours. The company manufacturing the bulbs claims that the average life of their bulbs is 1600 hours. Is the claim acceptable at 5% level of significance? The value of t at 5% level of significance and 24 degree of freedom is 1.711.	Co4	1.2.2 2.8.4	U, A																				
3	The mean life time of sample of 100 fluorescent light bulbs produced by a company is computed to be 1570 hours with a standard deviation of 120 hours. The company claims that the average life of the bulbs produced by it is 1600 hours. Using the level of significance of 0.05, is the claim acceptable?	Co4	1.2.2 2.8.4	U, A																				
4	Children having one parent of blood type M and the other type N will always be one of the three types M, MN, N and average proportion of these will be 1 : 2 : 1. Out of 300 children of above parents, 30% are of group M, 45% of group MN and rest are of group N. Find the Chi-square value and examine the correspondence between the theory and the experiment.	Co4	1.2.2 2.8.4	U, A																				
5	Test the hypothesis at 5% level of significance that the presence or absence of hypertension (HT) is independent of smoking habits from the following data of 180 persons. <table><tr><td></td><td>Non smokers</td><td>Moderate smokers</td><td>Heavy smokers</td></tr><tr><td>HT</td><td>21</td><td>36</td><td>30</td></tr><tr><td>No HT</td><td>48</td><td>26</td><td>19</td></tr></table>		Non smokers	Moderate smokers	Heavy smokers	HT	21	36	30	No HT	48	26	19	Co4	1.2.2 2.8.4	U, A								
	Non smokers	Moderate smokers	Heavy smokers																					
HT	21	36	30																					
No HT	48	26	19																					
6	There are two different choices to stimulate a certain chemical process. To Test whether the variance of the yield is the same no matter which catalyst is used. A sample of 10 batches is produced using the first catalyst and of 11 using the second. If the resulting data is $s_1^2 = 0.14$ and $s_2^2 = 0.28$,test the hypothesis of equal variance at 5% level. (F-Test)	Co4	1.2.2 2.8.4	U, A																				
7	For the following two independent samples of any problem, test <table><tr><td>X</td><td>15.0</td><td>8.0</td><td>3.8</td><td>6.4</td><td>27.4</td><td>19.0</td><td>35.3</td><td>13.6</td><td></td></tr><tr><td>y</td><td>18.8</td><td>23.1</td><td>10.3</td><td>8.0</td><td>18.0</td><td>10.2</td><td>15.2</td><td>19.0</td><td>20.2</td></tr></table> the equality of variance at 10% level of significance. (Using F-Test)	X	15.0	8.0	3.8	6.4	27.4	19.0	35.3	13.6		y	18.8	23.1	10.3	8.0	18.0	10.2	15.2	19.0	20.2	Co4	1.2.2 2.8.4	U, A
X	15.0	8.0	3.8	6.4	27.4	19.0	35.3	13.6																
y	18.8	23.1	10.3	8.0	18.0	10.2	15.2	19.0	20.2															
8	Two random samples gave the following data: Size Mean Variance Sample I 8 9.6 1.2 Sample II 11 16.5 2.5 Can we	Co4	1.2.2 2.8.4	U, A																				

	conclude that the two samples have been drawn from the same normal population? $F_{0.05, (10,7)} = 3.64$.																													
9	12 dice were thrown 4096 times and a throw of 6 was reckoned as a success; the observed frequencies were given below : <table border="1"><tr><td>No. of successes :</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7 & above</td></tr><tr><td>Frequencies</td><td>447</td><td>1145</td><td>1181</td><td>796</td><td>380</td><td>115</td><td>24</td><td>8</td></tr></table> Find the value of chi-square on the basis of the hypothesis that the dice were unbiased and hence show that the data are consistent with the hypothesis so far as the χ^2 – test is concerned. The value of χ^2 for 7 degrees of freedom at 5% level of significance is 14.067.									No. of successes :	0	1	2	3	4	5	6	7 & above	Frequencies	447	1145	1181	796	380	115	24	8	Co4	1.2.2 2.8.4	U, A
No. of successes :	0	1	2	3	4	5	6	7 & above																						
Frequencies	447	1145	1181	796	380	115	24	8																						
10	The following table gives the number of aircraft accidents that occurred during the various days of the week. Test whether the accidents are uniformly distributed over the week: <table border="1"><tr><td>Days :</td><td>Mon</td><td>Tue</td><td>Wed</td><td>Thu</td><td>Fri</td><td>Sat</td></tr><tr><td>No. of accidents</td><td>14</td><td>18</td><td>12</td><td>11</td><td>15</td><td>14</td></tr></table>									Days :	Mon	Tue	Wed	Thu	Fri	Sat	No. of accidents	14	18	12	11	15	14	Co4	1.2.2 2.8.4	U, A				
Days :	Mon	Tue	Wed	Thu	Fri	Sat																								
No. of accidents	14	18	12	11	15	14																								
11	Fit a binomial distribution for the following data showing the survey of 800 families with 4 children and test the goodness of fit. <table><tr><td>No. of boys</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>No. of girls</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr><tr><td>No. of families</td><td>32</td><td>178</td><td>290</td><td>238</td><td>64</td></tr></table>									No. of boys	0	1	2	3	4	No. of girls	4	3	2	1	0	No. of families	32	178	290	238	64	Co4	1.2.2 2.8.4	U , A
No. of boys	0	1	2	3	4																									
No. of girls	4	3	2	1	0																									
No. of families	32	178	290	238	64																									

B.T. Level: R: Remembering, U: Understanding, A: Applying, N: Analyzing, E: Evaluating, C: Creating

Date:	
Assignment No: 10	
Topic:	Curve fitting by the numerical method
Sub Topics:	Least square Method and Fitting of straight lines, second degree parabola and various curves
Relevant CO:	Apply the least square method for curve fitting.
Objectives:	1. Understand Curve fitting by the numerical method. 2. To fit a straight line by method of least square. 3. To fit a Second degree parabola by method of least square. 4. To fit general curves by method of least square.

Sr. No.	Question	CO	PI	B.T. level																		
1	Find the best-fit values of a and b so that $y = a + bx$ fits the data given in the table. <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>y</td><td>1</td><td>1.8</td><td>3.3</td><td>4.5</td><td>6.3</td></tr></table>	x	0	1	2	3	4	y	1	1.8	3.3	4.5	6.3	Co5	1.2.1	U, A						
x	0	1	2	3	4																	
y	1	1.8	3.3	4.5	6.3																	
2	Apply the method of Least squares, to fit straight line $y = a + bx$ to the following set of observations. Also, estimate the value of y at $x=72$. <table><tr><td>x</td><td>65</td><td>66</td><td>67</td><td>67</td><td>68</td><td>69</td><td>71</td><td>73</td></tr><tr><td>y</td><td>67</td><td>68</td><td>64</td><td>68</td><td>72</td><td>70</td><td>69</td><td>70</td></tr></table>	x	65	66	67	67	68	69	71	73	y	67	68	64	68	72	70	69	70	Co5	1.2.1	U, A
x	65	66	67	67	68	69	71	73														
y	67	68	64	68	72	70	69	70														
3	Fit a straight line $y = a + bx$ for the following data : <table><tr><td>x :</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td></tr></table>	x :	1	2	3	4	5	6	7	8	9	Co5	1.2.1	U, A								
x :	1	2	3	4	5	6	7	8	9													

	<table><tr><td>y :</td><td>2</td><td>6</td><td>7</td><td>8</td><td>10</td><td>11</td><td>11</td><td>10</td><td>9</td></tr></table> Also find difference between the actual value of y and the value of y obtained from the fitted curve in each case when x = 5.	y :	2	6	7	8	10	11	11	10	9									
y :	2	6	7	8	10	11	11	10	9											
4	In a tensile of a metal bar, the following observations were made; where x represent the load in tones and y the elongation in thousandth of millimeters: <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>y</td><td>35</td><td>68</td><td>100</td><td>138</td><td>170</td></tr></table> Determine a linear relation expressing y in terms of x by the method of least squares	x	1	2	3	4	5	y	35	68	100	138	170	Co5	1.2.1	U, A				
x	1	2	3	4	5															
y	35	68	100	138	170															
5	Explain the method of least squares. Apply the method of Least squares to fit straight line to the following set of observations <table><tr><td>X</td><td>1</td><td>1.5</td><td>2</td><td>2.5</td><td>3</td><td>3.5</td><td>4</td></tr><tr><td>Y</td><td>1.1</td><td>1.3</td><td>1.6</td><td>2</td><td>2.7</td><td>3.4</td><td>4.1</td></tr></table>	X	1	1.5	2	2.5	3	3.5	4	Y	1.1	1.3	1.6	2	2.7	3.4	4.1	Co5	1.2.1	U, A
X	1	1.5	2	2.5	3	3.5	4													
Y	1.1	1.3	1.6	2	2.7	3.4	4.1													
6	Find the best-fit values of a, b and c so that $y = a + bx + cx^2$ fits the data given in the table. <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>y</td><td>35</td><td>68</td><td>100</td><td>138</td><td>170</td></tr></table>	x	1	2	3	4	5	y	35	68	100	138	170	Co5	1.2.1	U, A				
x	1	2	3	4	5															
y	35	68	100	138	170															
7	Apply the method of Least squares to fit the second degree parabola using the least square method to the following data. Also, estimate y at x=6. <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>y</td><td>5</td><td>12</td><td>26</td><td>60</td><td>97</td></tr></table>	x	1	2	3	4	5	y	5	12	26	60	97	Co5	1.2.1	U, A				
x	1	2	3	4	5															
y	5	12	26	60	97															
8	Determine the constants a and b by the least-square method such that $y = ae^{bx}$ fits the following data: <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>y</td><td>1</td><td>1.8</td><td>1.3</td><td>2.5</td><td>6.3</td></tr></table>	x	0	1	2	3	4	y	1	1.8	1.3	2.5	6.3	Co5	1.2.1	U, A				
x	0	1	2	3	4															
y	1	1.8	1.3	2.5	6.3															
9	Apply the method of Least squares, to fit a function of the form $y = ax^b$ <table><tr><td>x</td><td>1</td><td>1.5</td><td>2.0</td><td>2.5</td><td>3.0</td><td>3.5</td><td>4.0</td></tr><tr><td>y</td><td>1.1</td><td>1.3</td><td>1.6</td><td>2.6</td><td>2.7</td><td>3.4</td><td>4.1</td></tr></table>	x	1	1.5	2.0	2.5	3.0	3.5	4.0	y	1.1	1.3	1.6	2.6	2.7	3.4	4.1	Co5	1.2.1	U, A
x	1	1.5	2.0	2.5	3.0	3.5	4.0													
y	1.1	1.3	1.6	2.6	2.7	3.4	4.1													
10	Apply the method of Least squares, to fit a function of the form $y = ab^x$ to the following data: <table><tr><td>x :</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>y :</td><td>1</td><td>1.0</td><td>1.3</td><td>2.5</td><td>6.3</td></tr></table>	x :	0	1	2	3	4	y :	1	1.0	1.3	2.5	6.3	Co5	1.2.1	U, A				
x :	0	1	2	3	4															
y :	1	1.0	1.3	2.5	6.3															

B.T. Level: R: Remembering, U: Understanding, A: Applying, N: Analyzing, E: Evaluating, C: Creating